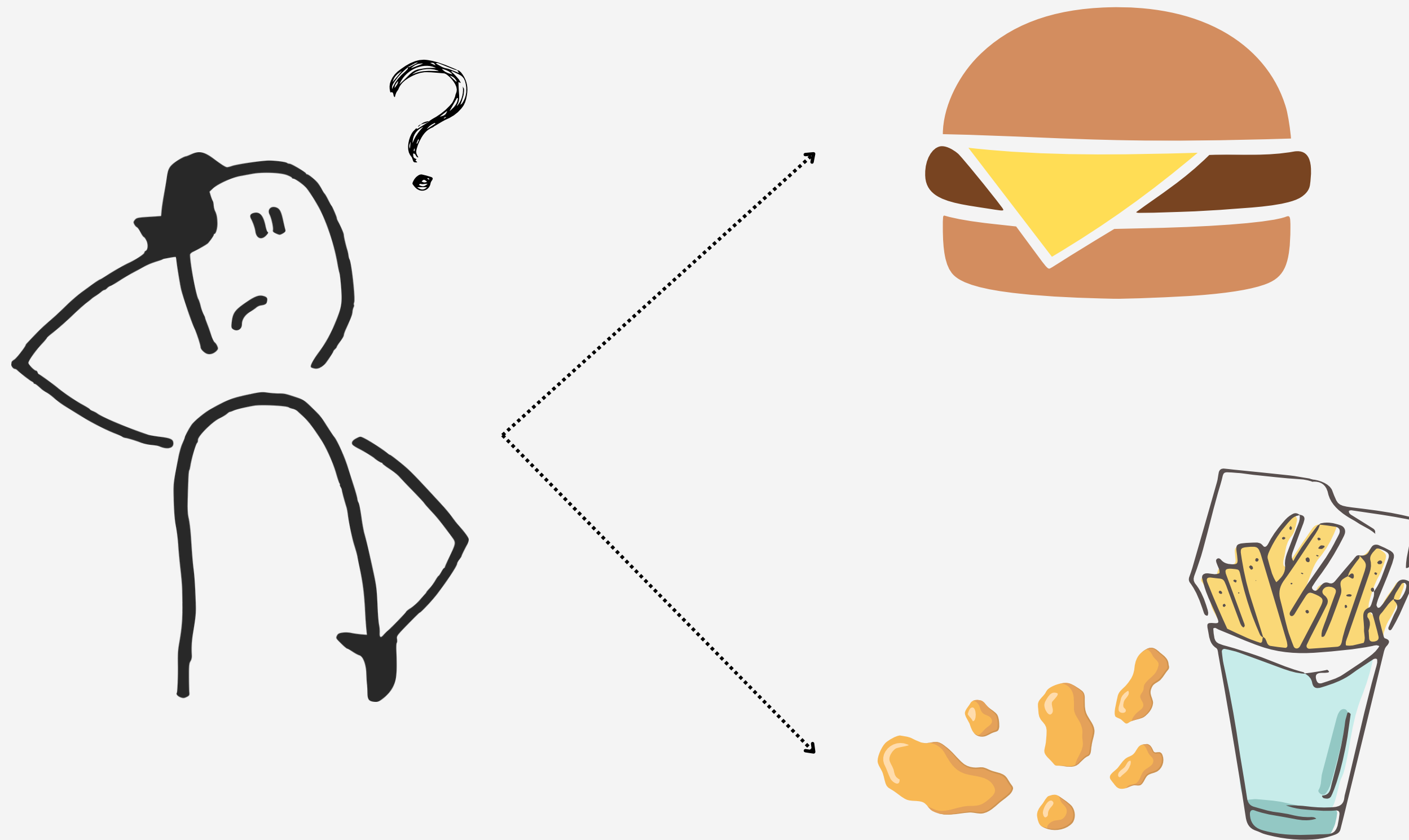


# WHAT IF THAT C.ELEGANS COULD ACTUALLY THINK ?



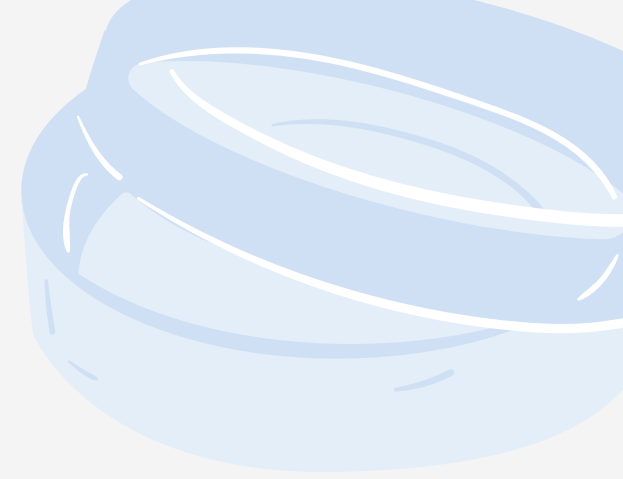


DO VERY SIMPLE ORGANISMS FOLLOW SIMILAR STRATEGIES WHEN THEY  
MOVE BETWEEN FOOD PATCHES ?

# BEHAVIORAL OPTIMIZATION: TESTING OPTIMAL FORAGING THEORY

**Nour Saad, Oceane Alvarez Benita Ibrahim M1 BBS**

**SUPERVISORS : Alfonso Pérez Escudero and Roger Lloret Cabot**



January- May 2026

# Contents

## 01 Introduction

- Biological question
- Theory

## 02 Methods

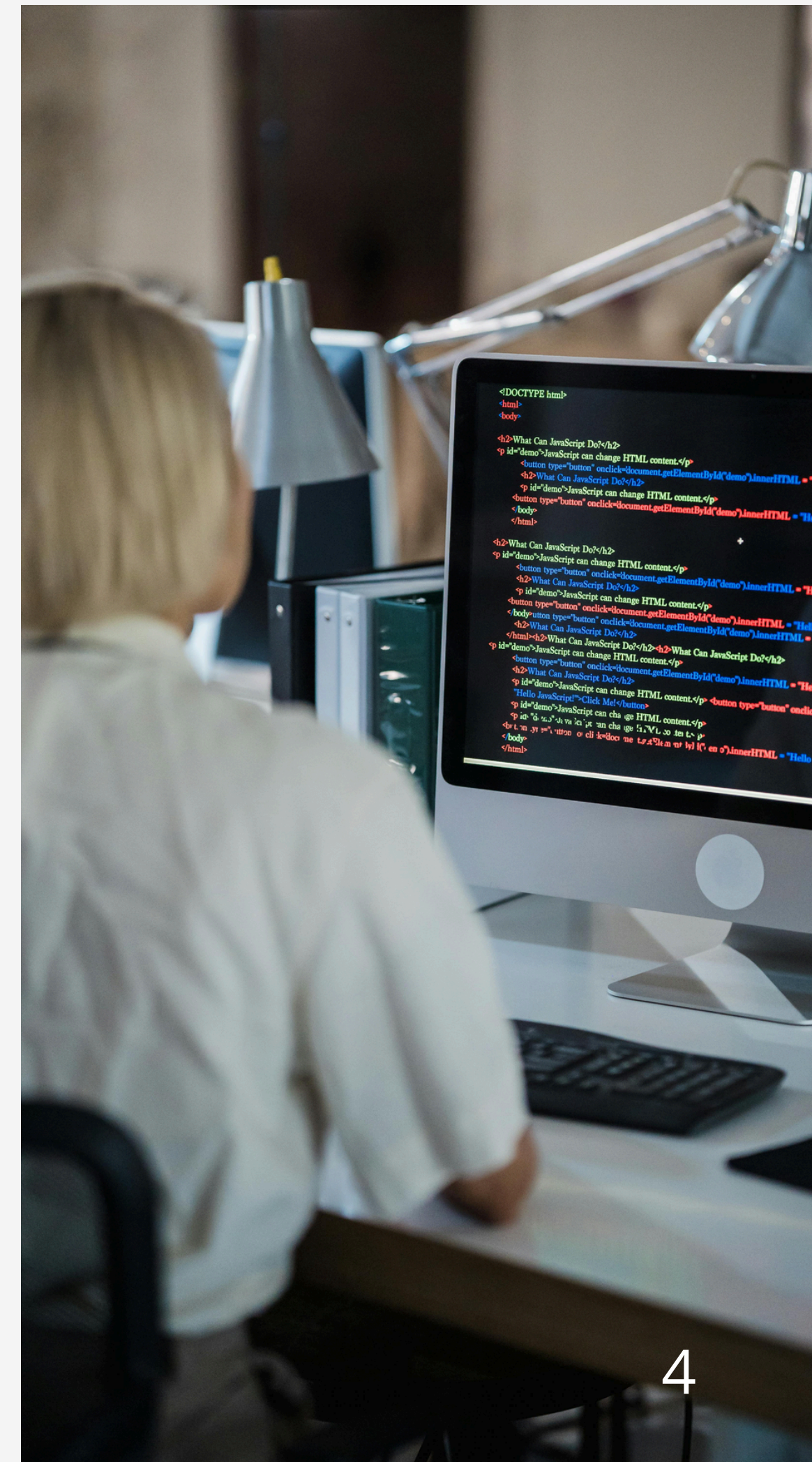
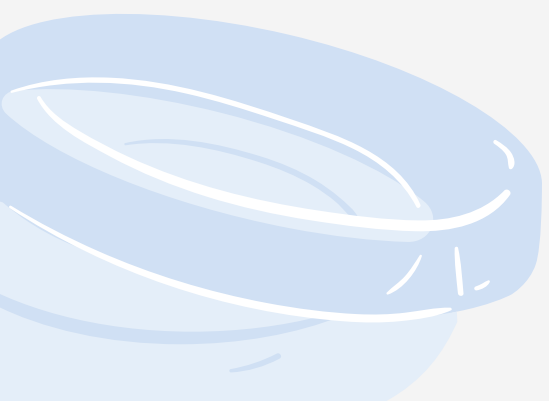
- Procedure
- Analysed the dataset

## 03 Results

- What we obtained ?

## 04 Discussion

- Interpretation





# Introduction

- **The Problem:** Why do organisms leave a patch of food?
- **Significance:** Understanding the mathematics of efficiency in the context of *C.elegans* worms.



# Methodology

**Phase 01**  
Overview of the dataset



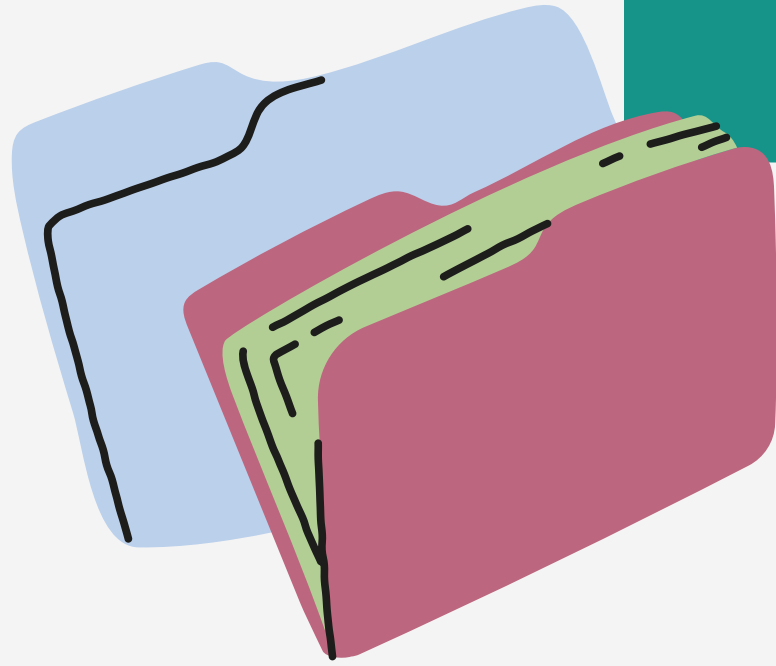
**Phase 02**  
Creation of a Clean Dataset

**Phase 03**  
Analysis of Multiple Worms

**Phase 04**  
Speed Analysis

# PHASE 01

## Overview of the dataset



### **Minipatches\_light\_20260116 :**

- containing experimental folders (that represents various videos)
- That contain many files



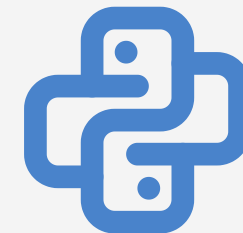
# PHASE 02

## Creation of a Clean Dataset



### inconsistent if:

- traj.csv is missing but traj.mat is not empty.
- foodpatches\_reviewed.mat is missing but traj.mat is not empty.



### A Python script :

- Checks file existence.
- Measures file size -> empty files.
- Labels each video as:  
OK  
NO WORM  
WEIRD

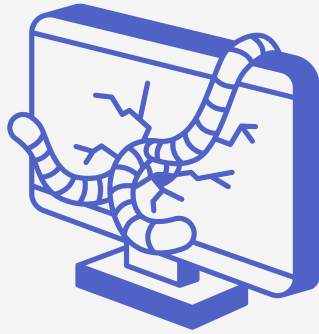




# PHASE 03

## Analysis of Multiple Worms

For each csv file:



Computation of  
the proportion of  
frames containing  
1+ worms

Manual  
inspection :  
ambiguous  
cases.



Assign labels:

- 0 worms
- 1 worm
- 1 worm with tracking errors
- 2+ worms



# PHASE 04

## Speed Analysis

For each csv file:

Computation of speed along trajectories

$$\text{Speed} = \sqrt{(xb - xa)^2 + (yb - ya)^2}$$

Plotting speed (y) over time (frames) (x)

To detect suspicious jumps => tracking errors/data corruption

Flag abnormal segments => exclusion/ correction

# RESULTS

# CSV files

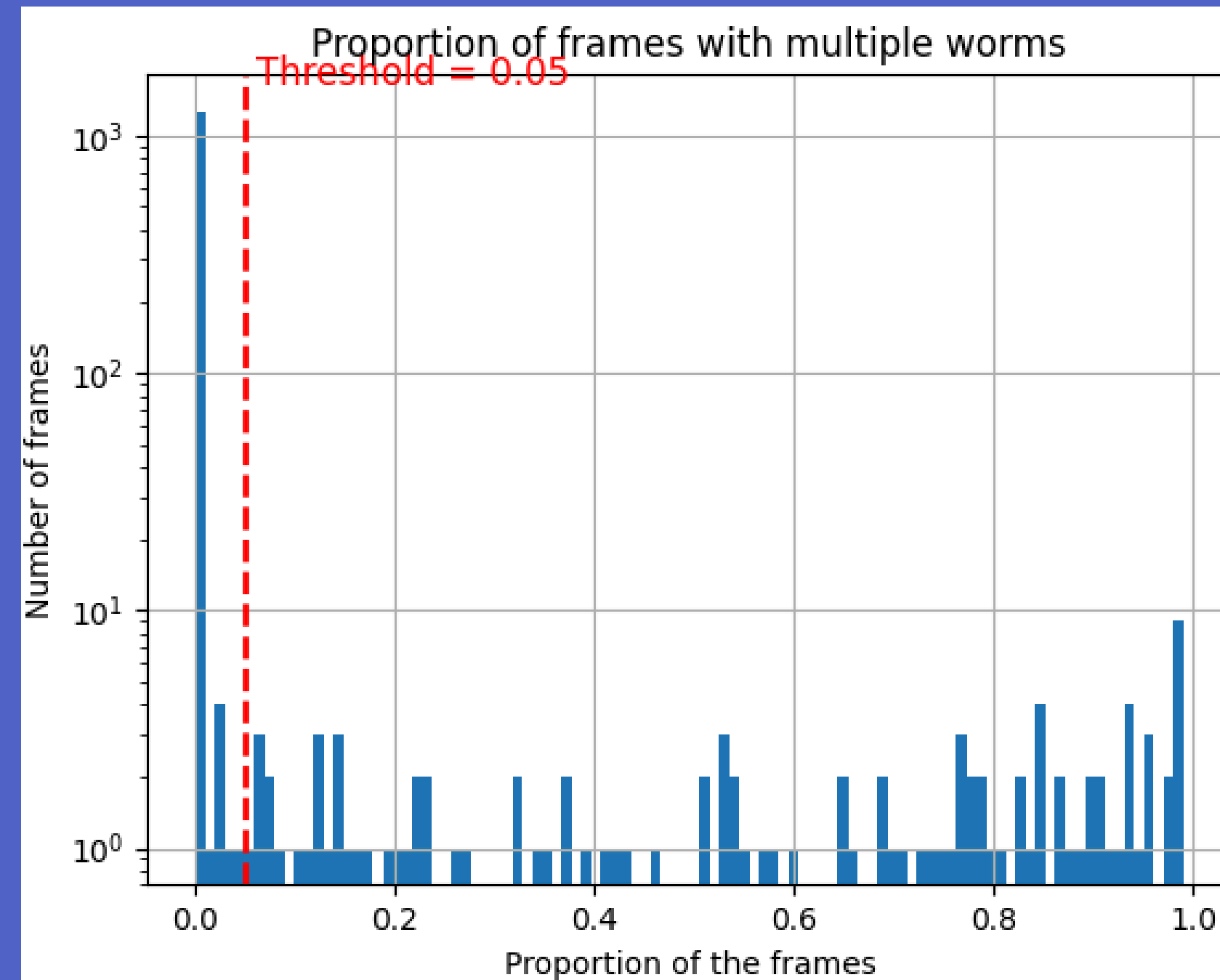
Phase 2 : Creation of a clean dataset

Phase 3 : Analysis of multiple worms

| Folder Path     | traj.csv | foodpat | traj.m | traj.mat empty | Diagnosis | 0 worms | 1 worm | 2+ worms  | proportion | final classification |
|-----------------|----------|---------|--------|----------------|-----------|---------|--------|-----------|------------|----------------------|
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 6255    | 27562  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 1588    | 33626  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 6662    | 4872   | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 3991    | 31033  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 0        | 0       | 1      | 1              | NO WORM   | 0       | 0      | 0         | 0.0        | 0 worms              |
| Documents/PTUT/ | 1        | 0       | 1      | 1              | NO WORM   | 293     | 1153   | 33511     | 0.9586348  | 2+ worms             |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 12135   | 22887  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 6159    | 28948  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 8919    | 25983  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 232     | 34680  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 1472    | 33629  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 0       | 1      | 1              | NO WORM   | 622     | 7732   | 26822     | 0.7625085  | 2+ worms             |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 3865    | 31358  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 323     | 34698  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 5966    | 29008  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 1        | 0       | 1      | 1              | NO WORM   | 16998   | 8147   | 139767756 | 13.9767756 | 1 worm with errors   |
| Documents/PTUT/ | 1        | 1       | 1      | 1              | NO WORM   | 13923   | 20675  | 0         | 0.0        | 1 worm clean         |
| Documents/PTUT/ | 0        | 0       | 1      | 1              | NO WORM   | 0       | 0      | 0         | 0.0        | 0 worms              |

# Histogram

## Phase 4: Analysis of Multiple Worms





# OUR TEAM

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.



Juliana Silva



Alexander Aronowitz

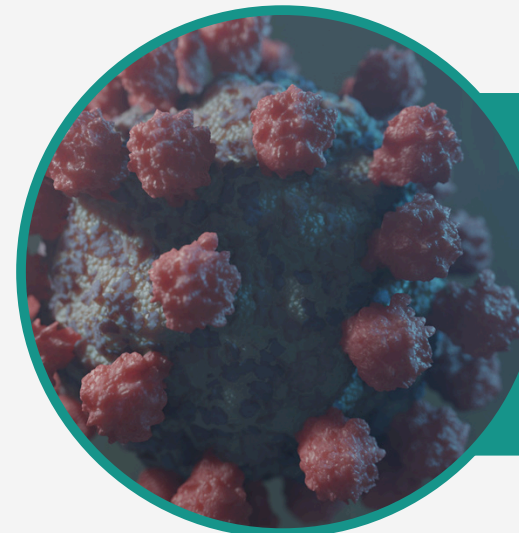


Olivia Wilson

# Conclusions

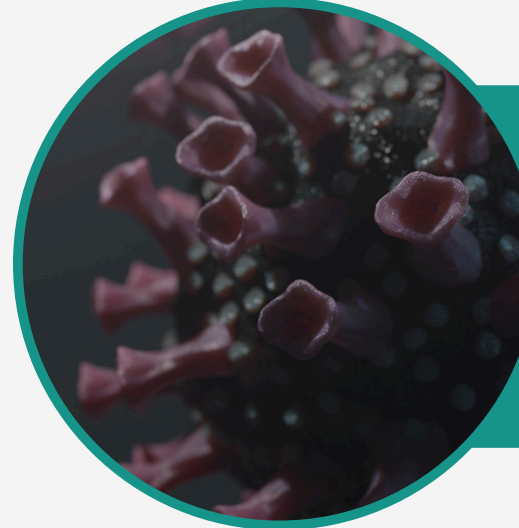
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.



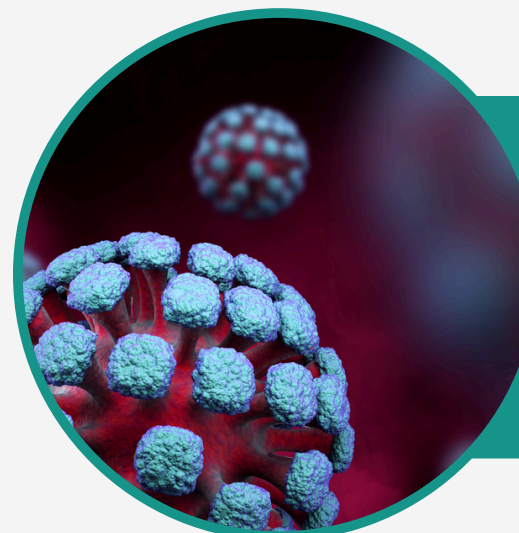
## Conclusion 01

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.



## Conclusion 02

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.



## Conclusion 03

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla ullamcorper nulla et blandit ultrices. Nulla aliquam congue enim. Duis imperdiet bibendum.



# THANK YOU

## Contact us



### Alexander Aronowitz

- +123-456-7890
- [www.reallygreatsite.com](http://www.reallygreatsite.com)
- [hello@reallygreatsite.com](mailto:hello@reallygreatsite.com)

# Resource page

